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D15C 57

Aim

To implement and apply **K-Means Clustering** and **Hierarchical Clustering** on the **Online Retail Customer Dataset** to identify different customer segments based on purchasing behavior.

Objective

The objectives of this experiment are as follows:

- To understand the concept of **unsupervised learning and clustering techniques**
 - To preprocess real-world transactional data for analysis
 - To perform **feature engineering using the RFM model (Recency, Frequency, Monetary)**
 - To apply **K-Means clustering algorithm** and analyze customer segmentation
 - To determine the optimal number of clusters using the **Elbow Method**
 - To apply **Hierarchical clustering** and interpret results using a dendrogram
 - To compare both clustering techniques and evaluate their effectiveness
 - To derive business insights from customer segmentation
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Dataset Source

- **Dataset Name:** Online Retail Dataset
 - **Source:**
<https://www.kaggle.com/datasets/joebeachcapital/customer-segmentation?resource=download>
 - **Domain:** E-commerce / Retail Analytics
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Dataset Description

The Online Retail dataset contains transaction-level data of a UK-based online store. Each row represents a single purchase transaction of a product by a customer.

Attributes in the Dataset

- **InvoiceNo:** Unique identifier for each transaction
 - **StockCode:** Unique product code
 - **Description:** Name of the product
 - **Quantity:** Number of items purchased in a transaction
 - **InvoiceDate:** Date and time of the transaction
 - **UnitPrice:** Price per unit of the product
 - **CustomerID:** Unique identifier for each customer
 - **Country:** Country of the customer
-

Need for Feature Engineering

The dataset is transactional in nature, which means it does not directly provide customer-level insights. Therefore, we transform the data into a customer-centric format using the **RFM model**, which is widely used in marketing analytics.

RFM Model Explanation

The RFM model evaluates customers based on three key factors:

1. Recency (R)

Recency measures how recently a customer has made a purchase. Customers who have purchased recently are more likely to engage again compared to those who have not purchased for a long time.

2. Frequency (F)

Frequency represents how often a customer makes purchases. A higher frequency indicates a loyal customer who regularly interacts with the business.

3. Monetary (M)

Monetary value indicates how much money a customer has spent. Customers with higher monetary values are considered more valuable to the business.

By combining these three features, we can effectively segment customers based on their behavior.

Algorithms Used

K-Means Clustering

K-Means is a partition-based clustering algorithm that divides the dataset into a predefined number of clusters (K). Each cluster is represented by its centroid, which is the mean of all data points in that cluster.

$$J = \sum_{j=1}^K \sum_{i=1}^n |x_i^j - c_j|^2$$

Working of K-Means

The algorithm follows an iterative process:

1. The number of clusters K is chosen initially
2. K centroids are randomly initialized
3. Each data point is assigned to the nearest centroid based on Euclidean distance
4. New centroids are calculated as the mean of the assigned points
5. Steps 3 and 4 are repeated until convergence

Key Concept: WCSS

WCSS (Within-Cluster Sum of Squares) measures the compactness of clusters. Lower WCSS indicates better clustering.

Elbow Method

The Elbow Method is used to determine the optimal number of clusters (K). A graph is plotted between the number of clusters and WCSS. The point where the graph starts to bend (forming an “elbow”) is considered the optimal value of K.

Hierarchical Clustering

Hierarchical clustering creates a tree-like structure called a dendrogram. In this experiment, we use **Agglomerative clustering**, which is a bottom-up approach.

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

Working of Hierarchical Clustering

1. Each data point starts as an individual cluster
2. The closest clusters are merged together
3. This process continues until all points form a single cluster

Ward Linkage

Ward linkage is used to minimize the variance within clusters during merging, resulting in more compact and meaningful clusters.

$$\Delta(A, B) = \frac{n_A n_B}{n_A + n_B} |m_A - m_B|^2$$

Limitations

K-Means Limitations

1. Fixed K: Requires specifying the number of clusters in advance.
2. Outlier Sensitivity: Centroids are affected by extreme values.
3. Spherical Assumption: Cannot detect non-spherical or irregular cluster shapes.

Hierarchical Clustering Limitations

1. Computational Complexity: Very slow for large datasets ($O(n^3)$ or $O(n^2 \log n)$).
2. Irreversibility: Once clusters are merged, they cannot be split.
3. Sensitivity to Noise: Prone to chaining effect and affected by noisy data.

Methodology

The following steps were followed to perform clustering:

Step 1: Data Loading

The dataset is loaded into the environment using pandas.

Step 2: Data Cleaning

Missing values, especially in CustomerID, are removed because clustering requires valid identifiers.

Step 3: Feature Creation

A new feature called TotalAmount is created using:

$$\text{TotalAmount} = \text{Quantity} \times \text{UnitPrice}$$

Step 4: RFM Table Creation

The dataset is grouped by CustomerID to calculate:

- Recency
- Frequency
- Monetary

Step 5: Data Scaling

StandardScaler is applied to normalize the data so that all features contribute equally.

Step 6: K-Means Clustering

K-Means algorithm is applied with different values of K.

Step 7: Elbow Method

WCSS values are plotted to determine optimal clusters.

Step 8: Hierarchical Clustering

Agglomerative clustering is applied to the dataset.

Step 9: Dendrogram

A dendrogram is plotted to visualize cluster formation.

Step 10: Visualization and Analysis

Clusters are analyzed and interpreted.

Code & Output

```
from google.colab import files
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import
StandardScaler
from sklearn.cluster import KMeans

# Upload file
uploaded = files.upload()

# Read dataset
df = pd.read_csv(list(uploaded.keys())[0])

# Select features
features = ['HP', 'Attack', 'Defense']
X = df[features]
```

```
# Scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# K-Means model
model = KMeans(n_clusters=3,
random_state=42)
df['Cluster'] = model.fit_predict(X_scaled)

# Visualization
sns.scatterplot(x=X['HP'], y=X['Attack'],
hue=df['Cluster'], palette='viridis')
plt.xlabel("HP")
plt.ylabel("Attack")
plt.title("K-Means Clustering")
plt.show()
```



from

```
google.colab import files
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import
StandardScaler
from sklearn.cluster import KMeans
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# Upload file
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# Read dataset
df = pd.read_csv(list(uploaded.keys())[0])
```

```
# Select features
features = ['HP', 'Attack', 'Defense']
X = df[features]
```

```
# Scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
# Elbow Method
```

```
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i,
                    random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)
```

```
# Plot Elbow Graph
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

```
# Optimal K
optimal_k = 3
```

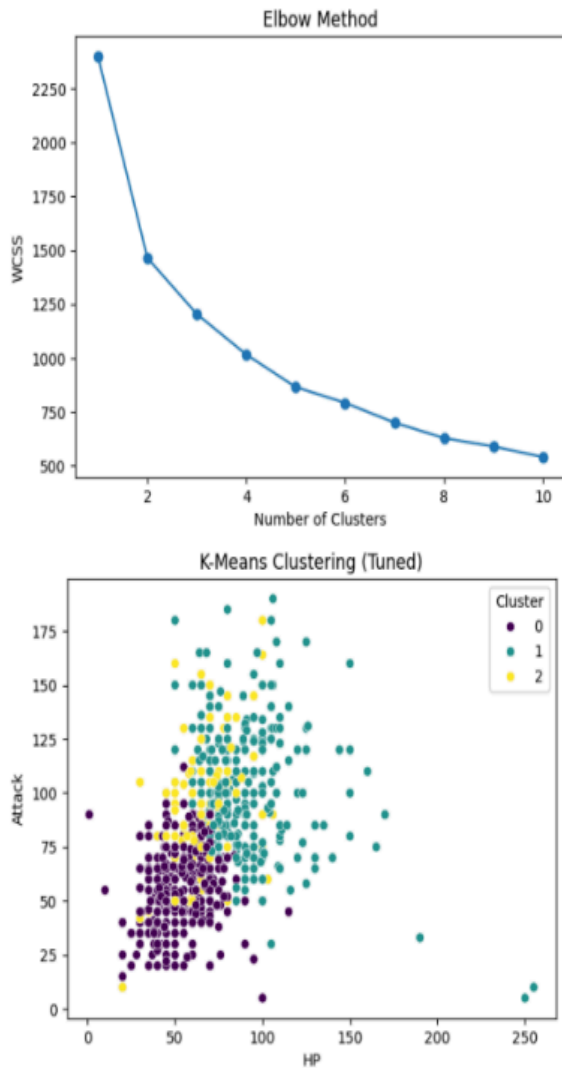
```
# Final K-Means
model = KMeans(n_clusters=optimal_k,
                random_state=42)
df['Cluster'] = model.fit_predict(X_scaled)
```

```
# Visualization
```

```

sns.scatterplot(x=X['HP'], y=X['Attack'],
hue=df['Cluster'], palette='viridis')
plt.xlabel("HP")
plt.ylabel("Attack")
plt.title("K-Means Clustering (Tuned)")
plt.show()

```



```

from google.colab import files
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import
StandardScaler
from sklearn.cluster import
AgglomerativeClustering

```

```

# Upload file
uploaded = files.upload()

# Read dataset
df = pd.read_csv(list(uploaded.keys())[0])

# Select features
features = ['HP', 'Attack', 'Defense']

```



```
X = df[features]
```

```
# Scaling
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

```
# Hierarchical Clustering
```

```
model =
```

```
AgglomerativeClustering(n_clusters=3)
```

```
df['Cluster'] = model.fit_predict(X_scaled)
```

```
# Visualization
```

```
sns.scatterplot(x=X['HP'], y=X['Attack'],
```

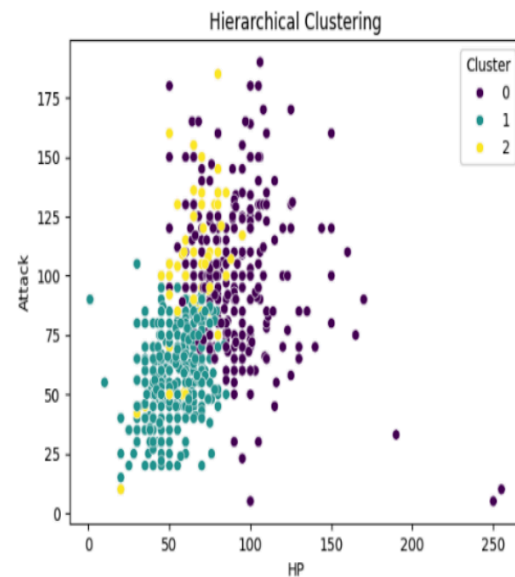
```
hue=df['Cluster'], palette='viridis')
```

```
plt.xlabel("HP")
```

```
plt.ylabel("Attack")
```

```
plt.title("Hierarchical Clustering")
```

```
plt.show()
```



```
from google.colab import files
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.cluster import AgglomerativeClustering
```

```
from scipy.cluster.hierarchy import dendrogram, linkage
```

```
# Upload file
```

```
uploaded = files.upload()
```

```
# Read dataset
```

```
df = pd.read_csv(list(uploaded.keys())[0])
```

```
# Select features
```

```
features = ['HP', 'Attack', 'Defense']
```

```
X = df[features]
```

```
# Scaling
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

```
# Dendrogram
```

```
plt.figure(figsize=(10, 6))
```

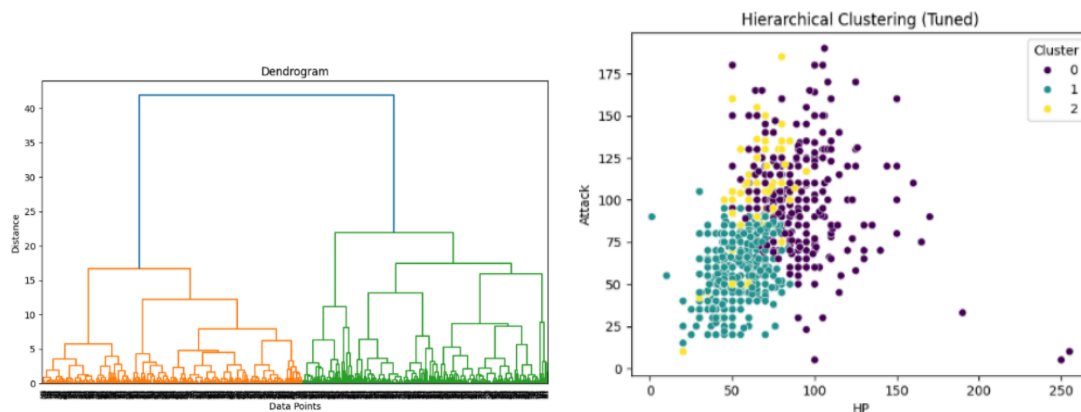
```

dendrogram(linkage(X_scaled, method='ward'))
plt.title("Dendrogram")
plt.xlabel("Data Points")
plt.ylabel("Distance")
plt.show()

# Hierarchical Clustering
model = AgglomerativeClustering(n_clusters=3)
df['Cluster'] = model.fit_predict(X_scaled)

# Visualization
sns.scatterplot(x=X['HP'], y=X['Attack'], hue=df['Cluster'], palette='viridis')
plt.xlabel("HP")
plt.ylabel("Attack")
plt.title("Hierarchical Clustering (Tuned)")
plt.show()

```



Results and Interpretation

Elbow Method Analysis

The Elbow graph shows a sharp decrease in WCSS up to $K = 3$, after which the decrease becomes gradual. This indicates that adding more clusters beyond 3 does not significantly improve the clustering. Therefore, the optimal number of clusters is selected as 3.

K-Means Clustering Interpretation

The K-Means algorithm divides customers into three distinct groups:

- The first cluster consists of customers who have high purchase frequency and high spending. These customers are considered premium or high-value customers.
- The second cluster represents customers with moderate purchasing behavior. These are regular customers who contribute steadily to revenue.
- The third cluster contains customers with low frequency and low spending. These are low-value or inactive customers.

This segmentation helps businesses identify which customers to target for marketing campaigns.

Dendrogram Analysis

The dendrogram visually represents how clusters are formed step by step. The vertical distance between merges indicates how different the clusters are. By drawing a horizontal line across the dendrogram at an appropriate height, we can identify the number of clusters.

In this case, the dendrogram suggests the presence of three major clusters, which aligns with the Elbow Method.

Hierarchical Clustering Interpretation

Hierarchical clustering produces clusters similar to K-Means, although slight differences may occur due to the different approach. It provides a clear visualization of how clusters are formed and merged, making it useful for understanding the structure of the data.

Comparison of Algorithms

K-Means is computationally efficient and works well for large datasets. However, it requires the number of clusters to be defined in advance. Hierarchical clustering, on the other hand, does not require predefined clusters and provides better interpretability through dendrograms, but it is computationally expensive.

Business Insights

The clustering results provide valuable business insights:

- High-value customers should be rewarded with loyalty programs and special offers
- Medium-value customers can be encouraged to spend more through targeted promotions
- Low-value customers can be re-engaged through discounts and personalized marketing

This segmentation helps businesses improve customer retention and maximize revenue.

Advantages

K-Means

- Simple and easy to implement
- Efficient for large datasets
- Provides clear cluster boundaries

Hierarchical Clustering

- No need to specify number of clusters initially
 - Provides dendrogram for visualization
 - Helps understand cluster relationships
-

Conclusion

In this experiment, K-Means and Hierarchical clustering techniques were successfully applied to the Online Retail dataset. By using the RFM model, customers were segmented into three meaningful groups based on their purchasing behavior.

The Elbow Method and dendrogram both indicated that three clusters were optimal. The results demonstrated that unsupervised learning techniques can effectively uncover hidden patterns in data and provide valuable business insights.

